

IPC Study Guide - First Two Topics

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1. IPC Communication and Types

What is IPC?

IPC (Inter-Process Communication) enables processes to exchange data and synchronize execution because each process has its own address space.

Why IPC?

Data sharing, synchronization, event notification, client-server communication, resource sharing.

Type	Use Case
Pipe	Parent-child byte stream
FIFO	Unrelated local processes
Message Queue	Structured messages
Shared Memory	Fast data sharing
Semaphore	Synchronization
Signals	Notifications
Unix/TCP Socket	Local/Network communication
mmap	Shared file memory

ASCII Diagram

Parent --> PIPE --> Child

C Example

```
pipe(fd); fork(); write(fd[1], msg,...); read(fd[0], buf,...);
```

Interview Tip: Shared memory is fastest but requires synchronization.

2. IPC Interview Questions and Answers

Q: What is IPC?

A: Mechanism for process communication and synchronization.

Q: Pipe vs FIFO?

A: Pipe is anonymous and for related processes; FIFO is named and supports unrelated processes.

Q: Fastest IPC?

A: Shared memory, because processes access the same memory region.

Q: Why semaphores?

A: Prevent race conditions in shared memory.

Q: System V vs POSIX?

A: POSIX APIs are newer, cleaner, and more portable.

Q: TCP vs Unix socket?

A: TCP works across networks; Unix sockets are local and faster.

Q: Signals limitation?

A: Suitable for notifications, not large data transfer.

Q: Blocking vs Non-blocking?

A: Blocking waits; non-blocking returns immediately.

Q: Deadlock causes?

A: Circular wait, inconsistent lock order, unreleased locks.

Notes:

- Close unused pipe ends.
- Always synchronize shared memory.
- Use Unix domain sockets for efficient local IPC.